Practice 1 Progress Report: Ark+ Classification on VinDr-CXR and ChestX-ray14

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Abstract

This report summarizes the current progress of Practice 1 using Ark+ with a Swin-B backbone for chest X-ray classification. The single-dataset VinDr-CXR experiment was completed for all 200 required cycles, reaching a final test mean area under the ROC curve (mAUC) of 0.9220 for the student and 0.9230 for the teacher. The ChestX-ray14 experiment reached 139 of 200 cycles before the presentation deadline, with the latest mAUC values of 0.8129 and 0.8125 for the student and teacher, respectively. Cyclic training on both datasets was also implemented and validated through 23 cycles, producing the required focused and unfocused measurements, but it was not completed because of time limits, memory failures, and distributed GPU allocation problems on ASU Sol. The experiments show rapid early improvement followed by a plateau, a slower convergence pattern for the larger ChestX-ray14 dataset, and preliminary evidence of cross-dataset transfer during cyclic training. The computational failures also identified practical improvements for Demo 1, especially shorter validation runs and more efficient evaluation during debugging.

1 Introduction

Practice 1 studies Ark+ for chest X-ray classification using a student–teacher learning framework and a Swin-B backbone. The first required experiment trains Ark+ independently on VinDr-CXR and ChestX-ray14 for 200 cycles. The second experiment trains one Ark+ model cyclically on both datasets and measures performance after each dataset update. For the cyclic experiment, performance measured on a dataset immediately after training on that same dataset is defined as *focused performance*; performance measured after training on the other dataset is defined as *unfocused performance*.

The main goals of the current work were to reproduce the single-dataset baselines, track the student and teacher performance trajectories, validate the cyclic training pipeline, and understand the practical computational requirements of the released Ark+ code on the ASU Sol cluster. Because the class GPU queue imposes strict allocation limits, an additional objective was to establish a reliable workflow for smoke testing, checkpointing, and resuming long experiments.

2 Hypotheses

Before training, three main behaviors were expected. First, the student model was expected to improve rapidly during the early cycles, while the exponential moving average teacher was expected to respond more slowly but become smoother and more stable later in training. Second, ChestX-ray14 was expected to require more time to converge than VinDr-CXR because it contains substantially more training images and more disease labels. Third, during cyclic training, performance on a dataset was expected to be strongest immediately after training on that dataset, while the unfocused measurement could reveal either forgetting or beneficial transfer from the other dataset.

3 Methods

3.1 Datasets and Tasks

VinDr-CXR contained 15,000 training images, 3,000 validation images, and 3,000 test images with six classification targets. ChestX-ray14 contained 75,312 training images, 11,212 validation images, and 25,596 test images with 14 classification targets. All images were processed at a resolution of 224×224 pixels as required by the assignment.

Task I(a) trained Ark+ only on VinDr-CXR. Task I(b) trained Ark+ only on ChestX-ray14. Task II trained one model cyclically on VinDr-CXR followed by ChestX-ray14 during each cycle and evaluated both datasets after each dataset update.

3.2 Model and Optimization

The released course Ark+ implementation was used with a Swin-B backbone initialized from ImageNet-pretrained weights. The configured global batch size was 200, optimization used stochastic gradient descent, and the learning-rate schedule used a 20-cycle warm-up. The student and teacher networks shared the same backbone structure, while the teacher parameters were updated by exponential moving average with a base momentum of 0.9. Training was performed in FP32.

For the single-dataset experiments, test-time augmentation was enabled using ten-crop evaluation. Mean AUROC across the dataset labels was used as the primary performance metric. A resumable checkpoint was saved after completed cycles,

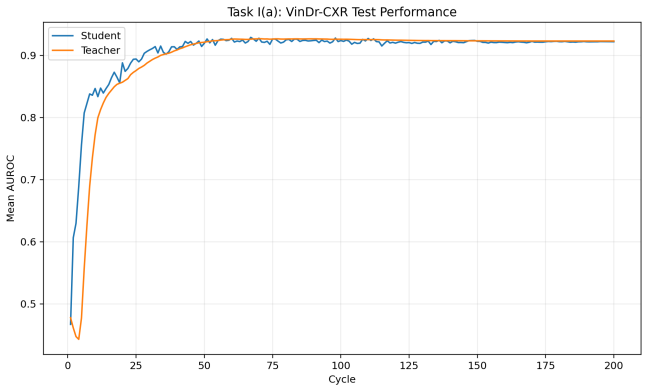


Figure 1: Task I(a) VinDr-CXR test mAUC for the Ark+ student and teacher across 200 cycles.

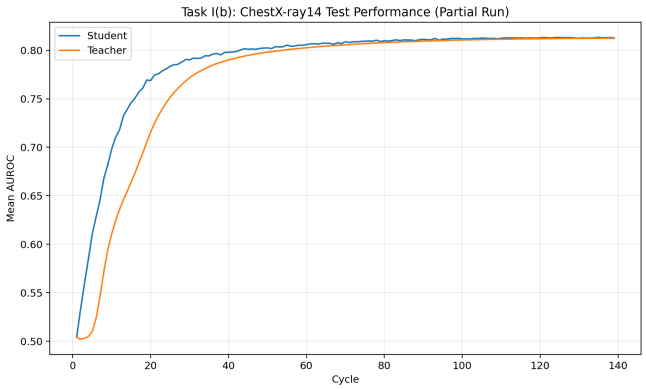


Figure 2: Task I(b) ChestX-ray14 test mAUC through the 139 completed cycles available before the presentation deadline.

allowing each experiment to continue across the four-hour class-QOS job limit.

3.3 SOL Workflow and Smoke Testing

Training was performed with the provided NVIDIA container on ASU Sol. Initial one-GPU smoke tests successfully initialized the datasets, Swin-B weights, student and teacher networks, and completed the first training cycle. However, evaluation was terminated by `SIGKILL`, showing that the one-GPU configuration was not sufficient for the default ten-crop evaluation workload. A subsequent four-A100 smoke test completed multiple train–evaluate cycles and validated the end-to-end pipeline.

The long experiments were therefore submitted as Slurm batch jobs and resumed from checkpoints after the four-hour allocation limit. This workflow successfully completed VinDr-CXR and made substantial progress on ChestX-ray14. For Task II, ten-crop evaluation produced severe overhead and memory pressure, so the cyclic experiment was restarted with center-crop evaluation for debugging and progress validation. Therefore, the Task II values reported here should be interpreted as preliminary pipeline results rather than a direct numerical comparison with the ten-crop Task I curves.

4 Results

4.1 Task I(a): VinDr-CXR

The VinDr-CXR experiment completed all 200 required cycles. The final test mAUC was 0.9220 for the student and 0.9230 for the teacher. The best student performance was 0.9289 at cycle 67, while the best teacher performance was 0.9268 at cycle 70.

The trajectory in Fig. 1 shows very rapid learning during the first part of training. Student mAUC increased from 0.4669 at cycle 1 to 0.8879 by cycle 20 and 0.9274 by cycle 60. After approximately 60–80 cycles, performance entered a plateau. The teacher initially lagged behind the student, but later became slightly more stable and finished marginally above the student.

Table 1: Current Task I performance summary.

Dataset	Cycles	Final S	Final T	Best S / T
VinDr-CXR	200/200	0.9220	0.9230	0.9289 / 0.9268
ChestX-ray14	139/200	0.8129	0.8125	0.8134 / 0.8125

4.2 Task I(b): ChestX-ray14

At the presentation deadline, ChestX-ray14 had completed 139 of the required 200 cycles. The latest available mAUC was 0.8129 for the student and 0.8125 for the teacher. The best student mAUC was 0.8134 at cycle 135, and the highest teacher mAUC observed so far was 0.8125 at cycle 139.

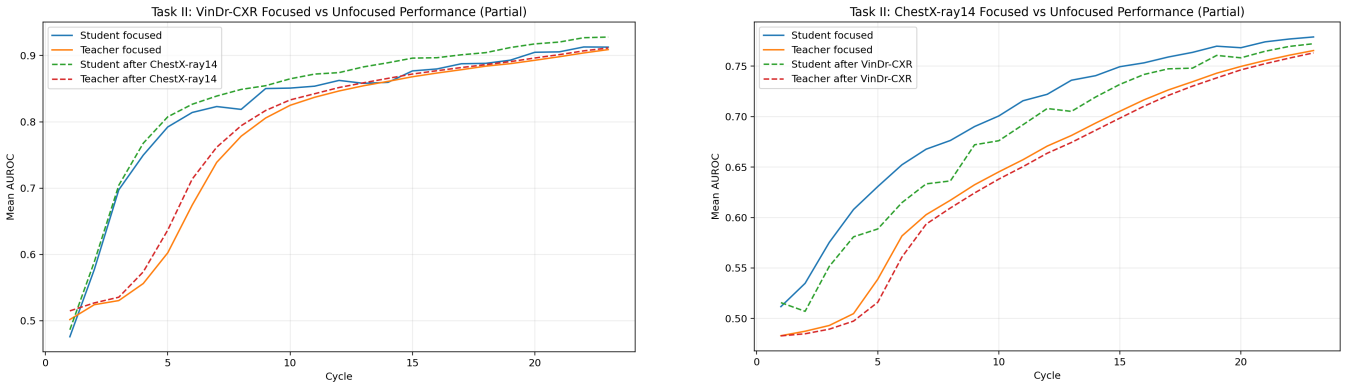
Figure 2 shows a slower and smoother improvement than the VinDr-CXR experiment. Student mAUC increased from 0.5043 at cycle 1 to 0.6983 at cycle 10, 0.7689 at cycle 20, and approximately 0.80 by cycle 50. From approximately cycle 100 onward, the student and teacher curves became very close, suggesting that the experiment was approaching a plateau even though the required 200 cycles had not yet been completed.

4.3 Task II: Preliminary Cyclic Training

Cyclic training was successfully implemented and produced valid focused and unfocused measurements through cycle 23 before the queued jobs later failed. Figure 3 shows the available trajectories for both datasets.

At cycle 23, VinDr-CXR student/teacher mAUC immediately after VinDr training was 0.9126/0.9089. After the subsequent ChestX-ray14 update, VinDr-CXR mAUC increased to 0.9278/0.9113. Across all 23 available cycles, the VinDr-CXR measurement after ChestX-ray14 training was consistently higher than the corresponding measurement immediately after VinDr-CXR training. This preliminary behavior suggests positive cross-dataset transfer in the early cyclic-training phase rather than immediate forgetting.

For ChestX-ray14 at cycle 23, mAUC after the VinDr-CXR update was 0.7722/0.7632 and increased to 0.7789/0.7655 after ChestX-ray14 training. This behavior agrees more closely with the expected focused/unfocused pattern: performance



(a) VinDr-CXR

(b) ChestX-ray14

Figure 3: Preliminary Task II cyclic-training mAUC. Solid curves represent focused performance and dashed curves represent performance after training on the other dataset. Results are available through cycle 23.

improved after training directly on ChestX-ray14. Because this experiment used center-crop evaluation after the ten-crop version proved too expensive, these values are used only to demonstrate the cyclic behavior and are not treated as final Task II results.

5 Discussion

5.1 Single-Dataset Behavior

The VinDr-CXR experiment supports the expected student–teacher behavior. The student adapted more quickly during the early cycles, while the teacher initially changed slowly because it is updated by exponential moving average. Once the model approached its performance plateau, the teacher became more stable and finished slightly above the student. The peak performance occurring around cycles 67–70, followed by a small decline toward cycle 200, also indicates that additional cycles did not produce continuous improvement.

ChestX-ray14 showed the same general progression but on a much slower timescale. This is consistent with its substantially larger training set and larger number of labels. Despite being incomplete, the trajectory already shows clear saturation: the student improved only modestly from approximately 0.8117 at cycle 100 to 0.8129 at cycle 139. The teacher continued to improve gradually and nearly matched the student by the latest cycle.

5.2 Cyclic Training and Transfer

The partial Task II results are useful because they verify that the implementation records both focused and unfocused performance after each dataset update. The early VinDr-CXR results were particularly interesting: training on ChestX-ray14 improved VinDr-CXR mAUC instead of reducing it. This suggests that, during the early training stage, the second dataset may reinforce shared chest-radiograph representations that also benefit VinDr-CXR. In contrast, ChestX-ray14 generally achieved its highest value after its own update, which is closer to the expected focused-performance behavior.

These observations are preliminary because only 23 cycles were completed and because Task II used center-crop evaluation after the original ten-crop configuration became computationally impractical. The full 200-cycle experiment is required before drawing strong conclusions regarding transfer, interference, or catastrophic forgetting.

6 Computational Limitations and Lessons Learned

The main limitation of this practice was not model implementation but computational execution under the available Sol resources. Several issues reduced the amount of training completed before the presentation deadline:

- Initial smoke tests were required to identify that one GPU could train successfully but failed during ten-crop evaluation.
- Four-GPU jobs were limited to approximately four hours and therefore required repeated checkpoint/resume cycles.
- Several ChestX-ray14 jobs terminated because of memory pressure, while others reached the Slurm time limit normally.
- The first Task II ten-crop run was terminated during evaluation. After switching the debugging run to center-crop evaluation, two long jobs successfully reached cycle 23.
- Multiple Task II jobs were queued automatically. Some later allocations exposed a distributed world size that was not compatible with the global batch size of 200, producing the error `Global batch size must be divisible by world size`. Because the jobs were chained, several dependent jobs failed consecutively before the problem was noticed, costing a substantial part of the available training window.

7 Planned Improvements for Demo 1

The following changes will be used to make the remaining experiments more reliable and efficient:

- Use short smoke tests before submitting long dependent job chains.
- During debugging, use center-crop evaluation or evaluate less frequently to reduce evaluation overhead. For the final reported comparison, use the same evaluation settings across all experiments.
- Continue ChestX-ray14 from its latest valid checkpoint until cycle 200.
- Resume Task II from a validated distributed configuration and complete the focused/unfocused trajectories before Demo 1.

These changes separate *debugging efficiency* from the *final evaluation*. Reducing evaluation frequency or test-time augmentation can help identify implementation problems quickly, while the final reported trajectories should use one consistent evaluation configuration.

8 Conclusion

Practice 1 successfully established a complete 200-cycle Ark+ baseline on VinDr-CXR and a substantial 139-cycle baseline on ChestX-ray14. VinDr-CXR reached a final mAUC of 0.9220/0.9230 for the student/teacher and showed a clear plateau after approximately 60–80 cycles. ChestX-ray14 reached 0.8129/0.8125 by cycle 139 and showed slower convergence consistent with its larger dataset and label space. Cyclic training was also implemented successfully and generated focused/unfocused measurements through cycle 23, with preliminary evidence of beneficial cross-dataset transfer for VinDr-CXR.

The incomplete experiments were primarily limited by queue time, four-hour job limits, evaluation memory cost, and distributed GPU allocation failures rather than by the Ark+ training code itself. These issues provided a clear debugging strategy for the next iteration. The remaining Task I(b) and Task II cycles will be completed and presented in Demo 1 using a more reliable long-training workflow and shorter validation runs before long job chains are submitted.